

VIGNESHWARAN C

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Professional Summary

Computer Science student at **St. Joseph's College of Engineering & IIT Madras** (BS Data Science) specializing in autonomous AI agent systems, multi-tenant SaaS architectures, and real-time backend pipelines. Proven track record architecting low-latency tool-calling orchestration engines, deterministic evaluation systems, and enterprise RBAC workflows across Python, FastAPI, React, Node.js, and PostgreSQL.

Education

St. Joseph's College of Engineering

B.E. in Computer Science and Engineering – CGPA: 8.13

Chennai, India

2024 – Present

IIT Madras

B.S. in Data Science and Applications (Dual Degree)

Online / Chennai, India

2024 – Present

Devi Academy Senior Secondary School

Class 12 (CBSE) – 92.2%

Chennai, India

2024

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Java, Dart, Bash

AI & Autonomous Agents: ReAct Agent Loops, Tool/Function Calling, Ollama, LangChain, MediaPipe, OpenCV, Scikit-learn

Backend & Systems: FastAPI, Node.js, Express.js, REST APIs, PostgreSQL, SQLite, Redis, Firebase Firestore, Linux IPC

Frontend & Tools: React, Next.js, Redux Toolkit, RTK Query, Flutter, Tailwind CSS, Docker, Git, Linux (Arch/Hyprland)

Professional Experience

Application Development Intern

ATRIBS Global Technology Solutions

Jun – Jul 2025 | Dec 2025 – Jan 2026

Chennai, India

- Selected for a second term based on performance; retained among top-performing interns.
- Engineered **Police Akka** (dispatch & citizen SOS app) with Firebase real-time streams, GPS tracking, and Groq LLM.
- Delivered architecture presentations on Industry 4.0 systems to enterprise mentors from Rockwell Automation.

AI Automation & Data Science Intern

Toeddro Adventures Pvt. Ltd.

Apr 2026

Remote

- Built automated ingestion pipelines structuring lead data across web sources; reduced manual operational workflows.

Featured Projects

J.A.R.V.I.S – Autonomous Desktop Agent & OS Engine | *Python, Ollama, Hyprland, OpenCV, Bash*

- Architected an autonomous OS agent running multi-step **ReAct loops** via local LLMs (Ollama) and function-calling schemas.
- Built an async tool engine with 15+ deterministic OS tools for workspace routing, process telemetry (*psutil*), and vision capture.
- Engineered continuous wake-word detection with noise calibration and synchronized execution states with desktop UI panels.

Law-Suite – Multi-Tenant Enterprise Legal SaaS | *React, Redux Toolkit, Node.js, Express, PostgreSQL*

- Designed a 14-module multi-tenant SaaS platform managing case lifecycles, automated billing, and document versioning.
- Implemented 4-tier hierarchical RBAC with middleware-level authorization checks and immutable audit logging.
- Built authentication infrastructure featuring JWT access/refresh token rotation, bcrypt hashing, and rate-limiting security.

NeuroMotion AI – Biomechanical Motion Analysis Engine | *Python, FastAPI, MediaPipe, DTW, Flutter*

- Engineered a temporal motion analysis pipeline tracking 33 spatial landmarks per frame to calculate 6 kinematic joint angles.
- Implemented Dynamic Time Warping (DTW) for speed-invariant sequence alignment against expert reference models.
- Built a deterministic rule engine (10 constraint rules) generating automated diagnostics and maximum-deviation heatmaps.

Certifications & Badges

- **Google Cloud:** Engineer AI Agents with Agent Development Kit (ADK) | MLOps for Generative AI
- **Microsoft:** Applied Skills: AI-Assisted Dev with GitHub Copilot | Generative AI in Azure PostgreSQL
- **Oracle & MongoDB:** Oracle Cloud Infrastructure 2025 Associate (OCI) | MongoDB Document Model & Basics
- **Academic & Professional:** Cloud Computing (NPTEL) | Python Advanced (HackerRank) | Cisco (Cybersecurity & IoT)

Achievements, Hackathons & Leadership

- **REC Titanium 2026: Winner** – Smart City Debug War Room
- **Startup TN & VIT Hackathons:** Finalist (Tourism Hackathon) | Prompt-a-thon Award (VIT x Google Gemini)
- **Competitive Coding:** 3rd Prize (Spark Sync, SJCE) | 4th Place (Codeathon, CIT) | Srinivasa Ramanujan Math Competition
- **Leadership:** Microsoft Learn Student Ambassador Candidate | Technical Lead, College Symposium (200+ attendees)